Computer Workshop Practice

Ground to Mastery — Complete Study Notes

ICT 121 | Year 1 Semester 2 | DCS

Tools · Safety · Electronics · Printers · Power · E-Waste

Coursework 40% (Test 10% + Assignment 10% + Practical 20%) | Final Exam 60%

UNIT 1: LABORATORY TOOLS & SAFETY

Workshop tools, safety gadgets, first aid, fire safety, workshop rules, electrical safety

1.1 — Computer Workshop Tools

A tool is a piece of instrument or equipment used to perform a particular activity. Tools are categorised into hand tools, diagnostic tools, cleaning tools, ESD protection equipment, networking tools, and support equipment — all used for installation, maintenance, troubleshooting, and repair.

Hand Tools

Term / Concept	Definition / Explanation
Flathead screwdriver	Turns slotted screws — various sizes needed for different components
Phillips (cross) screwdriver	Turns cross-head screws — most common in computers and peripherals
Torx screwdriver	Star-shaped — used in newer laptops, Macs, and some drives
Needle-nose pliers	Grips small components, picks up jumpers and small screws in tight spaces
Wire cutters/strippers	Cuts and strips network cables or power cables — used in cabling work
Tweezers	Handles very small electronic parts that fingers cannot grip safely

ESD Protection Equipment

Term / Concept	Definition / Explanation
ESD wrist strap	Anti-static wrist strap — clips to ground — continuously drains static from body — most important ESD tool
Anti-static mat	Work surface that dissipates static electricity — connect wrist strap to mat
Anti-static bag	Pink/silver metallic bags — always store components in these when not installed
ESD-safe clothing	Special conductive clothing — used in professional clean rooms

Diagnostic Tools

Term / Concept	Definition / Explanation
Digital Multimeter (DMM)	Measures voltage, current, resistance — tests PSU, cables, components. Standard diagnostic tool for all technicians

Term / Concept	Definition / Explanation
POST diagnostic card	Plugs into motherboard — shows numeric error codes on startup display
Cable tester (LAN)	Tests network cables for continuity and correct wiring of all 8 wires
Oscilloscope	Displays digital and analogue signals accurately — analyses purity and timing of signals
Windows Performance Monitor	perfmon.msc — tracks detailed performance counters, logs system activity, identifies bottlenecks
Windows Resource Monitor	resmon — real-time CPU, memory, disk, network usage with process details
Windows Memory Diagnostic	mdsched — built-in tool to test RAM for errors
CHKDSK	Command-line tool — scans and repairs file system errors on drives
DirectX Diagnostic Tool	dxdiag — checks graphics, sound, and DirectX components

Cleaning Tools

Term / Concept	Definition / Explanation
Compressed air can	Blows dust out of fans, heatsinks, keyboards — never use mouth (moisture)
Contact cleaning chemicals	Cleans electrical contacts on connectors and circuit boards
Bristle brush	Removes stubborn dust from fans and heatsink fins
Hand vacuum cleaner	Sucks up loose dust — use anti-static variant near components

Network Tools

Term / Concept	Definition / Explanation
Crimping tool	Attaches RJ-45 connectors to network cables — essential for cable making
Punch-down tool	Connects wires to patch panels or network jacks in structured cabling
LAN cable tester	Checks wiring in Ethernet cables — verifies all 8 wires are connected correctly

Support Equipment

Term / Concept	Definition / Explanation
Soldering iron and solder	Repairs electronic circuits — joins wires and

Term / Concept	Definition / Explanation
	components to boards
Desoldering pump	Removes solder from circuit boards when replacing components
Magnifying lamp	Helps view small components and SMD parts clearly
Thermal paste	Heat-conducting compound between CPU and heatsink — fills microscopic gaps
Torch/flashlight	Illuminates inside dark cases during maintenance
Parts organiser tray	Holds screws and small parts during assembly — prevents loss
Tool kit box	Stores and organises all maintenance tools securely

💡 **EXAM TIP**
Know at least 10 tools and their functions. Know which tools are used for ESD, diagnostics, networking, and cleaning. A common exam question: "Name 5 tools in a computer workshop and their uses."

1.2 — Safety Rules in the Workshop

- Always wear an ESD wrist strap when handling components — static electricity destroys chips silently
- Power off AND unplug computer before opening case — never work on live equipment
- Hold circuit boards by edges only — never touch chips, solder points, or gold connectors
- Keep liquids away from equipment — even small drops cause short circuits and corrosion
- Keep workspace clean and organised — loose screws can short the motherboard
- Use correct screwdriver size — wrong size strips screw heads making removal impossible
- Store components in anti-static bags when not in use
- Never force connectors — if it does not fit, check orientation before forcing
- Students with health problems (epilepsy, medication) must report to workshop staff
- No food or drink in the workshop at any time
- Wear correct protective equipment for tools being used
- Never work alone — always have someone nearby in case of emergency
- Immediately notify workshop supervisor of any faulty or broken equipment

Electrical Safety

Domestic power supplies use 240V — sufficient to cause serious electric shock. CRT monitors generate and store up to 30,000 volts (30 kV) — dangerous even when switched off for some time. Always follow basic electrical safety guidelines when working on electrical equipment.

- Use chip extraction/insertion tools to handle chips — never bend pins
- Always replace blown fuses with one of the correct rating — check existing fuse was rated correctly
- Never work alone — always have someone nearby for emergencies

1.3 — First Aid

Term / Concept	Definition / Explanation
Electric shock	Do NOT touch victim while connected to power — disconnect power source first — call for help — start CPR if not breathing
Burns	Cool with running cold water for 10+ minutes — do NOT use ice, butter, or toothpaste
Eye injury	Flush with clean water for 15+ minutes — seek medical attention immediately
Cuts	Apply pressure with clean cloth — elevate limb if possible — seek help for deep cuts
Fire	Use correct extinguisher — evacuate if large fire — do NOT use water on electrical fires

1.4 — Fire Extinguisher Types

Term / Concept	Definition / Explanation
Class A (Water/Foam)	Wood, paper, textiles — NOT for electrical fires — water conducts electricity
Class B (CO2/Dry powder)	Flammable liquids — petrol, oil, solvents
Class C (CO2/Dry powder)	ELECTRICAL fires — computers use this type — CO2 leaves no residue and does not damage equipment
Class D	Metal fires — magnesium, titanium, sodium
CO2 extinguisher	Best for computer rooms — clean — no residue — no equipment damage — preferred choice

⚠️ **CRITICAL WARNING**
NEVER use water or foam extinguishers on electrical fires — water conducts electricity and causes electrocution. Always use CO2 on computer equipment.

UNIT 2: COMPUTER SYSTEM USAGE & ACCESSORIES

Connecting, powering, basic maintenance, and energy configuration

2.1 — Connecting a Computer System

1. Place monitor at comfortable eye level and appropriate viewing distance
2. Connect VGA/HDMI/DisplayPort cable from monitor to PC graphics port
3. Connect keyboard and mouse (USB or PS/2)
4. Connect network cable (RJ-45 Ethernet) to NIC port
5. Connect power cables to monitor and PC — verify voltage (240V in Uganda)
6. Power on monitor first, then press PC power button

2.2 — Basic Maintenance Tasks

Term / Concept	Definition / Explanation
Antivirus scan	Run full system scan regularly — update virus definitions before scanning for best results
Disk cleanup	Remove temporary files, empty Recycle Bin, clear browser cache — frees disk space
Software updates	Apply OS and application security patches regularly — prevents vulnerabilities
Defragmentation	HDDs only — run monthly — reorganises fragmented files for faster access. NEVER defragment SSDs
Backup	Regular backups to external drive or cloud — follow 3-2-1 rule (3 copies, 2 media types, 1 offsite)
Dust removal	Open case every 6 months — blow out dust with compressed air from fans, heatsinks, vents

Computer Maintenance Plan Steps

- Checking computer log files for errors and security alerts
- Assessing hard disk space — ensure sufficient free space
- Examining computer folder permissions — security check
- Monitoring computer temperature — prevent overheating
- Ensuring adequate redundancy of computer systems
- Examining security features and firewall settings
- Installing security software patches and updates
- Updating antivirus software on all computers
- Updating critical service packs and software updates
- Performing regular comprehensive backups

2.3 — Types of Maintenance

Term / Concept	Definition / Explanation
Preventive maintenance	Prevents possible failures before they occur — regular cleaning, updates, inspections — lengthens component life
Corrective maintenance	Fixes a problem that has already occurred — finding root cause, replacing faulty parts, repairs
Predictive maintenance	Uses diagnostic tools to monitor system health and predict failures before they happen — monitors CPU temp, battery levels, disk health

2.4 — Energy Star Configuration

- Enable sleep/hibernate after inactivity — reduces power consumption significantly
- Set monitor to turn off after 10-15 minutes idle
- Enable power management settings in BIOS/UEFI
- Unplug devices not in use — standby mode still consumes power

UNIT 3: ELECTRONICS FUNDAMENTALS FOR MAINTENANCE

Electrostatics, electronic components, colour codes, multimeters, component testing

3.1 — Electronic Components Overview

Electronic components are the elements of an electronic circuit that work together to create electronic systems. Computers are made of electronic components referred to collectively as hardware. Components are classified as either passive or active.

Passive Components

Term / Concept	Definition / Explanation
Characteristics	Do not require external power source. Cannot amplify signals. Consume, store, or dissipate energy. Bidirectional (current flows both ways)
Resistor (R)	Opposes current flow — dissipates energy as heat. Ohm's Law: V = IR. Value measured in Ohms (Ω)
Capacitor (C)	Stores energy in electric field — blocks DC, passes AC. Value measured in Farads (F). Q = C x V
Inductor (L)	Stores energy in magnetic field — opposes change in current. Value measured in Henry (H). Works on Faraday's law
Transformer	Transfers energy magnetically between circuits — no power gain — basis of all power grids
Fuse	Sacrificial protection — breaks circuit when current exceeds rating — one-time use

Active Components

Term / Concept	Definition / Explanation
Characteristics	Require external power source. Can amplify, control, switch, or generate signals. Usually unidirectional
Diode	Allows current in one direction only (rectification) — LEDs emit light. Controls current direction
Transistor (BJT, MOSFET)	Amplifies or switches signals — core of modern electronics — billions on a single chip
Integrated Circuit (IC)	Contains many transistors — op-amps, microcontrollers, logic gates, timers
SCR / Thyristor	Controlled switching for high-power applications

📌 **NOTE ON DIODES**

Term / Concept	Definition / Explanation
Some diodes (rectifier diodes) are sometimes grouped as passive, but most modern classifications place diodes under active components because they control current direction and involve non-linear behaviour.	

3.2 — Electrostatic Discharge (ESD)

Static electricity builds up on your body from everyday movement. When you touch a component, it discharges through the component — potentially destroying it instantly. Just 20 volts can damage a chip that runs on 3.3V or 5V. You may feel absolutely nothing, but the chip is permanently damaged.

Term / Concept	Definition / Explanation
ESD wrist strap	Continuously drains static from body to ground — most important tool in the workshop
Anti-static bag	Pink/silver metallic bags — always store components in these when not installed in computer
Anti-static mat	Work surface that drains static — connect wrist strap to mat for full protection
Humidity	Low humidity increases static risk — humid environments naturally safer for electronics
ESD-safe clothing	Special conductive clothing — used in professional electronics clean rooms

3.3 — Resistor Colour Code

Resistors use coloured bands to indicate their value. This is the IEC 60062 / EIA standard. Read bands from left to right — the tolerance band (Gold or Silver) is usually on the right, often separated by a wider gap.

Band Formats

- 4-band (most common): Digit 1 — Digit 2 — Multiplier — Tolerance
- 5-band (precision): Digit 1 — Digit 2 — Digit 3 — Multiplier — Tolerance
- 6-band (rare): adds Temperature Coefficient (TCR) as 6th band

Colour Band	Digit / Meaning
Black	0
Brown	1
Red	2
Orange	3
Yellow	4
Green	5

Colour Band	Digit / Meaning
Blue	6
Violet	7
Grey	8
White	9
Gold (tolerance)	±5%
Silver (tolerance)	±10%

💡 **COLOUR CODE MNEMONIC**
Memory Aid: BB ROY Great Britain Very Good Wife Black Brown Red Orange Yellow Green Blue Violet Grey White = 0 1 2 3 4 5 6 7 8 9

Resistor Reading Examples

- Brown–Black–Red–Gold → 1, 0, x100, ±5% = 1,000Ω (1 kΩ) ±5%
- Red–Violet–Orange–Gold → 2, 7, x1000, ±5% = 27 kΩ ±5%
- Yellow–Violet–Green–Silver → 4, 7, x100k, ±10% = 4.7 MΩ ±10%
- 5-band: Brown–Red–Black–Brown–Brown → 1, 2, 0, x10, ±1% = 1.2 kΩ ±1%

How to Read Resistor Colour Code (Procedure)

7. Hold resistor with tolerance band (Gold/Silver) on your RIGHT
8. Read colours from LEFT to RIGHT — note each colour
9. Use colour table to find digit values for bands 1 and 2 (or 1,2,3 for 5-band)
10. The multiplier band = 10 raised to the power of that colour's digit
11. Tolerance band (Gold = ±5%, Silver = ±10%) shows accuracy range

3.4 — Capacitor Colour Coding

Modern capacitors mostly use printed numbers (e.g., "104" = 100 nF = 10 x 10^4 pF). Older ceramic disc/tubular types use colour bands similar to resistors — value is in picofarads (pF).

- Bands 1-2 (or 1-3) = significant digits
- Next band = multiplier
- Next band = tolerance
- Sometimes last band = voltage rating or temperature coefficient
- Example: Brown–Black–Orange = 10 x 1,000 = 10,000 pF = 10 nF

3.5 — Inductor Colour Coding

Less common today (most are printed or SMD), but colour-ring inductors follow almost the same system as resistors — value is in microhenries (µH).

- Example: Red–Violet–Gold = 27 x 0.1 = 2.7 µH Tolerance ±10% (Silver)

3.6 — Using a Digital Multimeter (DMM)

A digital multimeter is a test tool used to measure voltage (volts), current (amps), and resistance (ohms). It is the standard diagnostic tool for electronics technicians. The face has: Display, Buttons, Dial (rotary switch), and Input jacks.

Multimeter Settings

Term / Concept	Definition / Explanation
Voltage (V) — DC	Set to DC (V⎓) — probes ACROSS component — black to COM, red to V terminal
Voltage (V) — AC	Set to AC (V~) — for mains power testing — same probe placement
Current (A)	Set to A range — meter placed IN SERIES with circuit (break circuit, insert meter)
Resistance (Ω)	Set to Ω range — POWER OFF component first — place probes across component
Continuity	Beeps when circuit is complete — tests cables, fuses, switches — power off first
Diode test	Tests diodes — shows voltage drop (~0.6V for normal silicon diode)

Measuring Voltage — Step by Step

12. Plug black test lead into COM terminal, red into V terminal
13. Move dial to V~ for AC voltage, or V⎓ for DC voltage
14. Set range higher than expected voltage — if too low, reading is inaccurate
15. Touch red probe to positive, black probe to negative of component or source
16. Read display — for battery: AA = ~1.5V, 9V battery = ~9V

Measuring Resistance — Step by Step

17. Insert black lead in COM, red lead in Ω terminal
18. POWER OFF the component being tested — always
19. Set dial to a number in the resistance (Ω) section — start high if unsure
20. Place probes across both ends of the resistor or component
21. Read resistance value on display

3.7 — Testing Basic Electronic Components

Term / Concept	Definition / Explanation
Fuse	Continuity setting — should BEEP if good — no beep means blown fuse — replace same rating
Switch	Continuity test — should BEEP when switched ON, silent when OFF
Cable / wire	Continuity test — beep = wire intact — no beep =

Term / Concept	Definition / Explanation
	broken wire somewhere
Battery	Voltage test DC — AA = ~1.5V, 9V battery = ~9V, car battery = ~12V
Resistor	Resistance (Ω) setting — compare reading to colour code value — within tolerance = good
PSU (Power Supply)	Use 20-24 pin ATX tester OR jump green wire to any black wire — fan should spin immediately
Bulb	Continuity test — should beep when filament is intact — no beep = blown

Practical Lab: Resistor Measurement Experiment

Objective: Identify resistor values using colour codes and verify with DMM.

22. Choose three different resistors randomly from the set
23. Read resistor value using the colour code method — record theoretical value
24. Set DMM to Ω (resistance) range appropriate for expected value
25. Power off, place probes across resistor terminals
26. Record measured value from DMM display
27. Compare theoretical vs measured — difference should be within tolerance (Gold = ±5%)

📋 **ASSIGNMENT EXAMPLE**

Assignment Example: Resistor Red–Violet–Orange–Gold 1. Set DMM to resistance (Ω) — 200kΩ range. Place probes on resistor leads. 2. Expected: 27 x 1000 = 27 kΩ ±5% (Range: 25.65 kΩ to 28.35 kΩ) 3. If DMM reads 27 kΩ — correct. If very different — resistor may be damaged or wrong.

💡 **EXAM TIP**

Know the full colour code table. Know how to read 4-band and 5-band resistors. Know how to use a multimeter for voltage, resistance, and continuity. Practice is essential — this unit has a practical component worth 20%.

UNIT 4: PRINTERS, SCANNERS & PHOTOCOPIERS

Types, operation, connectivity, troubleshooting, maintenance

4.1 — Printers Overview

A printer is an output device that produces text and graphics on a physical medium such as paper. Printed information is called hard copy because it exists physically. Printers are grouped into impact and non-impact types.

- Impact printer: forms characters by striking a mechanism against an ink ribbon that contacts paper
- Non-impact printer: forms characters without striking the paper directly

Printing speed is measured in pages per minute (ppm) or characters per second (cps) for dot-matrix. Printer resolution is measured in dots per inch (dpi) — higher dpi = better quality.

4.2 — Types of Printers

Term / Concept	Definition / Explanation
Daisy Wheel Printer	Impact printer — uses a rotating wheel print head — hammer strikes spoke against paper. Advantage: letter-quality output. Disadvantage: very slow, cannot print graphics
Dot-Matrix Printer	Impact printer — print head with pins strikes ribbon. 9-pin or 24-pin (24-pin better quality). Prints multipart forms. Noisy. Speed measured in cps. Can handle dusty environments
Line Printer	High-speed impact — prints entire line at once. Speed in lines per minute (lpm). Used with mainframes and networks in manufacturing/shipping
Inkjet Printer	Non-impact — sprays tiny ink droplets. Colour capable. Quiet. Expensive ink cartridges. Ink may smear on ordinary paper. Speed in ppm
Laser Printer	Non-impact — uses toner powder + heat + drum. Very fast (4-30+ ppm). High quality 600-1200 dpi. Uses electrostatic process. Expensive toner but cheap per page
Thermal Printer	Heat-sensitive paper — receipt printers, label printers — no ink or toner needed
3D Printer	Builds 3D objects layer by layer — plastic filament or resin
Plotter	Moves pen over paper — used for engineering drawings — precise lines

How a Laser Printer Works

28. Document sent to printer — stored as full page before printing

29. Laser beam "draws" the document on a selenium-coated drum using electrical charges
30. Drum is rolled in toner (dry powder) — toner adheres to charged areas only
31. Toner transferred onto paper
32. Heat and pressure fuse toner permanently to paper (fuser unit)

Laser printers can be multifunction peripherals (MFP/MFC) — printing, scanning, photocopying, and faxing in one device.

How an Inkjet Printer Works

33. Print head moves across paper on a carriage
34. Tiny nozzles spray precise ink droplets onto paper
35. Each colour (Cyan, Magenta, Yellow, Black — CMYK) applied in passes
36. Paper advances between passes until page is complete

Colour Printing

Laser printers use black-only or CMYK toner (Cyan, Magenta, Yellow, Black — separate cartridges). Two types:

- Four-pass printers: carousel rotates — applies each colour one at a time — slower
- One-pass printers: four separate lasers and drums — all colours applied at once — faster

4.3 — Printer Connectivity

Term / Concept	Definition / Explanation
USB	Most common — direct cable connection — plug and play — for single PC use
Network (wired Ethernet)	RJ-45 cable — shared printer on LAN — needs static IP address for reliability
WiFi	Wireless network connection — convenient — connects via router — for home/office sharing
Bluetooth	Short range wireless — mainly for mobile printing from phones
Parallel port (LPT)	Old 25-pin standard — mostly obsolete — found only on very old equipment

4.4 — Factors When Buying a Printer

- Cost of hardware (initial purchase price)
- Laser or inkjet (running costs differ significantly)
- Print speed (ppm)
- Duplex printing (automatic double-sided saves paper)
- Print quality (dpi)
- Paper types supported
- Warm up time (laser printers need warm-up)
- Reliability and warranty

4.5 — Printer Problems & Troubleshooting

Term / Concept	Definition / Explanation
No output	Check power, cable, driver installation, print queue — clear any stuck jobs in queue
Poor print quality	Low ink/toner, dirty print head, wrong paper type — run head cleaning cycle
Paper jam	Open ALL access doors, remove paper GENTLY without tearing, check rollers for debris
Streaks on output	Dirty drum (laser printer) or clogged nozzle (inkjet) — clean or replace component
Printer offline	Check connection, restart print spooler service, check IP address (network printers)
Slow printing	Draft mode setting, network congestion, insufficient printer memory — upgrade RAM or change settings

4.6 — Scanner Types & Operation

Term / Concept	Definition / Explanation
Flatbed scanner	Glass surface with lid — highest quality — best for photos, books, and bound documents
Sheet-fed scanner	Paper fed through automatically — faster, higher volume — cannot scan books or bound items
Handheld scanner	Portable — user drags over document — lower quality — convenient for field use
Drum scanner	Professional publishing quality — highest resolution — used in print industry

Scanner Operation

Term / Concept	Definition / Explanation
Pre-scan	Quick low-resolution preview before final scan — use to adjust crop and settings
Resolution (DPI)	Dots per inch — higher DPI = better quality but larger file size — 300 DPI minimum for documents, 600+ for photos
Interface cable	USB most common — or network connection for shared scanner
Scanning to network	Configure scanner IP address — share via network for multiple users

4.7 — Photocopiers

Term / Concept	Definition / Explanation
Multifunctional (MFP)	Combines printing, scanning, copying, and faxing — easy internet connectivity for sharing
Desk printer/photocopier	Compact unit for small offices
Wide format	Prints large documents — engineering drawings, banners, posters
Label printers	Specialist — prints adhesive labels — thermal or inkjet technology

💡 **EXAM TIP**
Know all printer types, how laser works (drum, toner, fuser), how inkjet works (nozzles, CMYK), connectivity types, and troubleshooting steps. Scanner DPI and pre-scan are common exam questions.

UNIT 5: POWER PACKS & UPS

Computer power supply unit, UPS types, power troubleshooting using multimeter

5.1 — Computer Power Supply Unit (PSU)

The PSU converts AC mains power (220-240V in Uganda) to the DC voltages required by computer components. It is a critical component — failure means the computer will not operate at all.

Term / Concept	Definition / Explanation
Function	Converts 220-240V AC mains to multiple DC voltages used by components
+12V rail	Powers CPU, GPU (graphics card), motors in drives — highest demand rail
+5V rail	Powers motherboard, older drives, USB ports
+3.3V rail	Powers RAM, modern components, PCI slots
-12V rail	Some serial (COM) ports — rarely used
Wattage	Total power output capacity — 400W to 1000W typical — match to system components
80 PLUS rating	Efficiency rating — Bronze, Silver, Gold, Platinum — higher = less heat wasted

PSU Testing with Multimeter

Term / Concept	Definition / Explanation
+12V test	Set DMM to DC voltage — red probe on yellow wire, black on any black (ground) — should read ~12V
+5V test	Red probe on red wire, black on ground — should read ~5V
+3.3V test	Red probe on orange wire, black on ground — should read ~3.3V
Paperclip test	Short green wire (PS_ON) to any black wire — fan should spin — confirms PSU is functional

5.2 — UPS (Uninterruptible Power Supply)

A UPS provides battery backup during power outages — giving time to save work and shut down properly. Especially important in Uganda where power is unstable.

Term / Concept	Definition / Explanation
Offline/Standby UPS	Switches to battery when mains fails — slight delay (few milliseconds) — cheapest type — basic protection

Term / Concept	Definition / Explanation
Line-interactive UPS	Regulates voltage fluctuations (surges and sags) without switching to battery — good for Uganda — recommended for computers
Online/Double-conversion UPS	Always running on battery — cleanest, most stable power — most expensive — for servers and critical systems
VA rating	Volt-Ampere — capacity of UPS — match to total wattage of equipment connected
Runtime	How long battery lasts on full charge — depends on load vs capacity

Troubleshooting Power Pack with Multimeter

37. Test wall socket — set DMM to AC voltage — should read ~220-240V in Uganda
38. Test PSU output voltages on yellow (+12V), red (+5V), orange (+3.3V) wires
39. Check fuse in PSU — continuity test — beep = good, no beep = blown
40. Test power cable continuity — beep on each wire = cable intact
41. Test UPS battery — check output voltage when mains disconnected

5.3 — Power Problem Troubleshooting

Term / Concept	Definition / Explanation
Computer will not turn on	Check wall socket, check PSU power switch at back, test PSU, check power button cable connection on motherboard
Random shutdowns	Overheating (clean all dust), failing PSU (test voltages), power fluctuations (add UPS)
Beeping on startup	POST beep codes — single short beep = normal boot. Multiple beeps = hardware error — count beeps and look up code
PSU failure signs	Will not power on, burning smell, fan not spinning, random reboots, components not receiving power
Surge/spike damage	Use surge protector or UPS — never plug sensitive equipment directly into wall socket

💡 **EXAM TIP**
Know all PSU voltage rails and their colours. Know the three UPS types and when to use each. For Uganda context, Line-interactive UPS is recommended due to power instability. Know PSU testing procedure using multimeter.

UNIT 6: E-WASTE MANAGEMENT

Electronic waste — definition, categories, effects on people and environment, management strategies

6.1 — What is E-Waste?

E-waste (electronic waste) refers to discarded electrical or electronic devices and equipment. This includes obsolete computers, phones, TVs, printers, and any electronic device at end of life. E-waste is the fastest-growing waste category globally.

6.2 — Categories of E-Waste

Term / Concept	Definition / Explanation
Large IT equipment	Servers, mainframes, large printers, photocopiers, mainframe computers
Small IT equipment	PCs, laptops, monitors, printers, scanners, keyboards, mice, calculators
Telecom equipment	Mobile phones, routers, switches, landlines, telephone exchanges
Consumer electronics	TVs, radios, cameras, game consoles, stereos, DVD players
Household appliances	Microwaves, washing machines, refrigerators, electric cookers
Lighting equipment	Fluorescent tubes, LED strips, energy-saving bulbs, CFL lights

6.3 — Effects of E-Waste on People and Environment

Hazardous Materials Found in E-Waste

Term / Concept	Definition / Explanation
Lead	Found in CRT monitors, solder, batteries — neurotoxin — damages brain and nervous system — affects children severely
Mercury	Found in LCD backlights, thermostats — accumulates in food chain — damages kidneys and brain
Cadmium	Found in rechargeable batteries, semiconductors — carcinogen — causes kidney damage
Brominated flame retardants	Used in circuit boards and plastic casings — releases toxic gases when burned — lung damage
Chromium (Hexavalent)	Used in metal coatings — carcinogen — skin and respiratory damage
Barium	Found in CRT tubes — muscle weakness, heart

Term / Concept	Definition / Explanation
	problems

Environmental Effects

- Soil contamination — toxic chemicals leach into ground — damages crops and pollutes groundwater
- Air pollution — burning e-waste releases toxic fumes — causes severe respiratory problems
- Water pollution — chemicals reach rivers and lakes — affects aquatic life and drinking water
- Food chain contamination — fish absorb mercury — humans who eat fish are affected

6.4 — E-Waste Management

Term / Concept	Definition / Explanation
Personal level	Reduce — buy durable, repairable equipment. Reuse — donate working equipment to schools. Recycle — give to certified e-waste recyclers only
Institutional level	Asset register for all IT equipment. Data wiping before disposal (destroy data). Contracts with certified recycling companies
Government level	E-waste legislation and regulations. Establish certified recycling centres. Regulate import of old/used electronics. Extended Producer Responsibility (EPR) laws

E-Waste Management Cycle

42. Collection — designated collection points for old electronics
43. Sorting — separate by type, hazardous vs non-hazardous
44. Disassembly — take apart devices to separate materials
45. Material recovery — extract valuable metals (gold, copper, aluminium)
46. Safe disposal — hazardous materials disposed of through licensed facilities

Advanced Maintenance Tasks (Unit 6 Practical Component)

- Run computer defragmenter — for HDD only — improves file access speed
- Create backup of files — to external drive or cloud storage
- Connect to internet and download utility tools and programme updates
- Clean drives using appropriate disk cleaning software

📌 **EXAM: THE 3 Rs OF E-WASTE**
The 3 Rs of E-Waste: REDUCE (buy less, buy durable), REUSE (donate, refurbish), RECYCLE (certified facilities only). Know at least 3 hazardous materials and their health effects.

💡 **EXAM TIP**
Exam: Know at least 3 hazardous materials in e-waste AND their specific health effects. Know the e-waste management cycle all 5 steps. Know management at personal, institutional, and government

levels. Know the 3Rs.

QUICK REFERENCE & EXAM PREPARATION

Assessment Breakdown

Component	Marks
Test	10% — Written test on theory
Assignment	10% — Take-home assignment or case study
Practical	20% — Hands-on lab assessment — tools, multimeter, assembly
Final Examination	60% — End of semester exam — all 6 units
TOTAL	100%

Key Facts to Memorise

Tools

- ESD wrist strap = most important tool for preventing static damage
- CO2 extinguisher = only safe type for computer/electrical fires
- Multimeter measures: Voltage (V), Current (A), Resistance (Ω), Continuity
- LAN cable tester = for network cable continuity
- Crimping tool = attaches RJ-45 to network cables

Electronics

- Passive components: Resistor, Capacitor, Inductor, Transformer, Fuse
- Active components: Diode, Transistor, Integrated Circuit (IC)
- Colour code mnemonic: BB ROY Great Britain Very Good Wife (0-9)
- Gold = ±5% tolerance, Silver = ±10% tolerance
- 4-band resistor: Digit1 – Digit2 – Multiplier – Tolerance
- Resistance test: POWER OFF first, then measure across component
- Current test: meter in SERIES with circuit

Printers

- Impact: Dot-matrix, Daisy wheel, Line printer
- Non-impact: Inkjet, Laser, Thermal, 3D
- Laser process: Laser → Drum → Toner → Paper → Fuser (heat)
- Inkjet: sprays CMYK ink droplets through nozzles
- Resolution = DPI (dots per inch) — 600-1200 DPI for laser
- MFP = Multifunction Printer (print + scan + copy + fax)

Power

- PSU converts 220-240V AC → DC (+12V, +5V, +3.3V)
- Yellow = +12V, Red = +5V, Orange = +3.3V, Black = Ground

- UPS types: Offline (cheapest), Line-interactive (best for Uganda), Online (most expensive)
- VA rating = capacity of UPS

E-Waste

- Lead → brain/nervous system damage (CRT monitors, solder)
- Mercury → kidneys/brain, food chain (LCD backlights)
- Cadmium → cancer, kidneys (rechargeable batteries)
- 3Rs: Reduce, Reuse, Recycle
- E-waste cycle: Collection → Sorting → Disassembly → Recovery → Disposal

Common Exam Questions & Model Answers

Q: Name 5 tools in a computer workshop and state their uses.

- ESD wrist strap — prevents static discharge from damaging sensitive components
- Phillips screwdriver — removes cross-head screws from computer cases and drives
- Multimeter — measures voltage, current, and resistance to test components
- Compressed air can — blows dust from fans, heatsinks, and keyboards
- Crimping tool — attaches RJ-45 connectors to network cables

Q: State the safety precautions when working in a computer workshop.

- Wear ESD wrist strap when handling components
- Power off and unplug computer before opening the case
- Hold boards by edges — never touch chips or solder points
- Keep liquids away from equipment
- Never work alone — keep someone nearby for emergencies
- Use CO2 extinguisher only for electrical fires

Q: Explain how a laser printer works.

A laser beam draws the document on a selenium-coated drum using electrical charges. The drum is rolled in toner (dry powder) which adheres to the charged areas. The toner is transferred to paper and permanently fused using heat and pressure from the fuser unit.

Q: What are the hazardous materials in e-waste and their effects?

- Lead — found in CRT monitors and solder — damages brain and nervous system
- Mercury — found in LCD backlights — damages kidneys and brain, accumulates in food chain
- Cadmium — found in rechargeable batteries — carcinogen, causes kidney damage

Q: Differentiate between impact and non-impact printers.

Impact printers form characters by physically striking a mechanism against an ink ribbon that contacts paper — examples include dot-matrix, daisy wheel, and line printers. Non-impact printers form characters without striking the paper — examples include inkjet (sprays ink droplets) and laser printers (uses toner and heat).